

Seat No.: \_\_\_\_\_

Enrolment No. \_\_\_\_\_

**NATIONAL FORENSIC SCIENCES UNIVERSITY**  
**M. Sc. Digital Forensics and Information Security - Semester - II -April-2026**

**Subject Code: CTMSDFIS24 SII P3**

**Date: 28/04/2026**

**Subject Name: Malware Analysis and Forensics**

**Time: 2.30 PM TO 5.30 PM**

**Total Marks: 100**

**Instructions:**

1. Write down each question on separate page.
2. Attempt all questions.
3. Make suitable assumptions wherever necessary.
4. Figures to the right indicate full marks.

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Marks</b> |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| <b>Q.1</b> | <b>Attempt any three.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |              |
| (a)        | What is the importance of malware analysis? What are the challenges in malware analysis?                                                                                                                                                                                                                                                                                                                                                                                                             | <b>08</b>    |
| (b)        | List and explain the different types of malware analysis techniques.                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>08</b>    |
| (c)        | Explain the PE File format in detail.                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>08</b>    |
| (d)        | Malware is designed with specific objectives in mind. Explain any four such goals of malware and discuss the types of malware that are commonly used to accomplish them.                                                                                                                                                                                                                                                                                                                             | <b>08</b>    |
| <b>Q.2</b> | <b>Attempt any three.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |              |
| (a)        | You have just received a suspicious executable suspected to be malware. Instead of running it and risking system compromise, you decide to investigate it safely using a disassembler.<br>What is a disassembler and how does it help in understanding such a malicious program? Describe the step-by-step approach you would follow to analyze this malware using a disassembler.<br>Finally, justify when using a disassembler is the best choice for analysis and when it may not be appropriate. | <b>08</b>    |
| (b)        | What is the difference between static linking and dynamic linking?<br>How can the libraries and functions in an executable be helpful in malware analysis.                                                                                                                                                                                                                                                                                                                                           | <b>08</b>    |
| (c)        | What are the indicators of packed malware? How will you unpack UPX packed malware using static and dynamic analysis tools.                                                                                                                                                                                                                                                                                                                                                                           | <b>08</b>    |
| (d)        | Explain the following malware techniques:<br>i) Process Hollowing<br>ii) Hook injection                                                                                                                                                                                                                                                                                                                                                                                                              | <b>08</b>    |

Q.3

Attempt any three.

- (a) Explain how will you acquire and analyse a Windows malware sample using memory forensic techniques. 08
- (b) How will you run a malware in a safe environment and use basic dynamic analysis tool to monitor them 08
- (c) Explain how will you perform advanced dynamic analysis using a debugger? 08
- (d) Explain the following: 08
- i) Source level debugging
  - ii) Assembly level debugging
  - iii) Kernel mode debugging
  - iv) User debuggging

Q.4

Attempt any two.

- (a) Consider the following Assembly code: *factorial* 07

```

_start:
mov eax, 1 eax = 1
mov ecx, [num] ecx = num
program_loop:
cmp ecx, 1 while (ecx > 1) {
jle done ecx = eax * ecx
imul eax, ecx ecx --
dec ecx } result = eax
jmp program_loop
done:
mov [result], eax

```

Explain each line of the above Assembly code. What is the Assembly code doing? Write an equivalent C construct for the given Assembly code.

- (b) Consider the following Assembly code:

```

_start:
mov eax, 10 eax = 10
mov ebx, 20 ebx = 20
cmp eax, ebx if (eax < ebx) {
jg part1 ecx = ebx
mov ecx, ebx else
jmp end_part ecx = eax
part1:
mov ecx, eax

```

*max of 2 numbers*

end\_part:  
; further instructions

Explain each line of the above Assembly code. What is the Assembly code doing? Write an equivalent C construct for the given Assembly code.

- (c) Consider the following Assembly code:
- ```
mov eax, 20 ;  
mov ecx, 4 ;  
xor edx, edx ;  
div ecx
```

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Explain each line of the above Assembly code. What is the final value in eax, edx, and ecx.

**Q.5**

**Attempt any two.**

- (a) Explain the techniques malware can implement to prevent from being analysed by various analysis tools.
- (b) What do you understand by 'Malware signature'? Why is it important in malware analysis. Explain how the analysis of strings in a malware executable be helpful?
- (c) How can you monitor network activities and registry activities by malware sample.

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**END OF PAPER**